

Experiment No.1

```
a = 10
```

```
b = 3
```

```
# Arithmetic Operators
```

```
print("Addition:", a + b)
```

```
print("Subtraction:", a - b)
```

```
print("Multiplication:", a * b)
```

```
print("Division:", a / b)
```

```
print("Modulus:", a % b)
```

```
# Comparison Operators
```

```
print("a == b:", a == b)
```

```
print("a != b:", a != b)
```

```
print("a > b:", a > b)
```

```
print("a < b:", a < b)
```

```
print("a >= b:", a >= b)
```

```
print("a <= b:", a <= b)
```

```
# Logical Operators
```

```
print("True and False:", True and False)
```

```
print("True or False:", True or False)
```

```
print("not True:", not True)
```

Assignment Operators

```
c = a
```

```
c += b
```

```
print("c += b:", c)
```

```
c -= b
```

```
print("c -= b:", c)
```

```
c *= b
```

```
print("c *= b:", c)
```

Bitwise Operators

```
print("a & b:", a & b)
```

```
print("a | b:", a | b)
```

```
print("a ^ b:", a ^ b)
```

```
print("~a:", ~a)
```

```
print("a << 1:", a << 1)
```

```
print("a >> 1:", a >> 1)
```

Membership Operators

```
list1 = [1, 2, 3, 4]
```

```
print("2 in list:", 2 in list1)
```

```
print("5 not in list:", 5 not in list1)
```

Identity Operators

```
x = [1, 2]
```

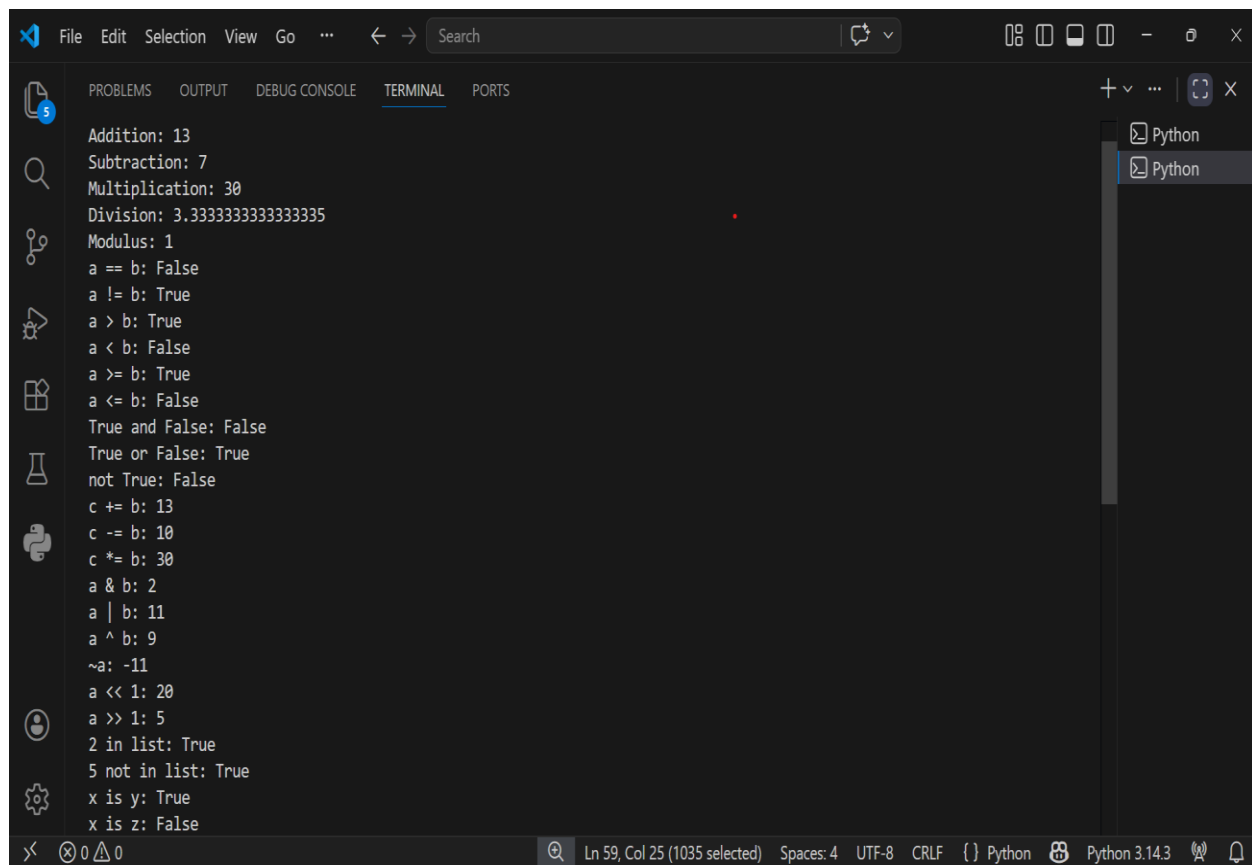
```
y = x
```

```
z = [1, 2]
```

```
print("x is y:", x is y)
```

```
print("x is z:", x is z)
```

Output:



The screenshot shows a Python IDE terminal window with the following output:

```
Addition: 13
Subtraction: 7
Multiplication: 30
Division: 3.3333333333333335
Modulus: 1
a == b: False
a != b: True
a > b: True
a < b: False
a >= b: True
a <= b: False
True and False: False
True or False: True
not True: False
c += b: 13
c -= b: 10
c *= b: 30
a & b: 2
a | b: 11
a ^ b: 9
~a: -11
a << 1: 20
a >> 1: 5
2 in list: True
5 not in list: True
x is y: True
x is z: False
```

The terminal window is titled "TERMINAL" and shows the output of the code. The status bar at the bottom indicates "Ln 59, Col 25 (1035 selected) Spaces: 4 UTF-8 CRLF {} Python Python 3.14.3".